

ECN 712 Fall 2025
Problem Set X, Due FRIDAY 21 November

1. Consider the “Vertical Integration and Welfare” problem from ASMT VII.
 - (a) Analyze it as an extensive form game with two players, the manufacturer and the retailer; in particular, identify the set of possible histories, each player’s information sets, and each player’s preference representation over *outcomes* (terminal histories).
 - (b) What is the strategy set of each player, and what is each player’s preference representation over *strategy profiles*?
 - (c) Was the solution to this problem on assignment VII a subgame-perfect Nash equilibrium (SPNE) *outcome*? Why or why not? If it was a SPNE outcome, then identify a Nash equilibrium that is not subgame perfect; if it was *not* a SPNE outcome, identify a SPNE. Explain clearly why the strategy profile you identify is a Nash equilibrium and why it is or is not subgame perfect.
2. MWG, Exercise 9.B.5, *omitting* the last clause of part (e), namely “or if we replace the phrase *pure-strategy Nash equilibrium* with the phrase *mixed-strategy Nash equilibrium*” [Correction to the question preamble: Action set M_i has k_i actions; m_i is a typical element of M_i as is implied in the body of the problem, not the number of elements in M_i as written in the preamble.]
3. *Entry*. Consider the following two-stage entry game with **two** firms. This problem examines the relationship between the nature of post-entry competition and SPNE outcomes.
 - In the first stage, the firms simultaneously decide whether or not to enter a market “ I ” for IN, “ O ” for OUT). If a firm chooses I , it pays a setup cost of $K > 0$; if it chooses O , then it has nothing further to choose and its profit is zero.
 - In the second stage, the firms observe the entry decisions made at the first stage; any firms that have entered play a simultaneous-move game, either Bertrand or Cournot. If only one firm enters, then of course that firm is a monopolist.

The only cost that a firm has is the setup cost $K > 0$ (incurred in the first stage if it chose I): the second stage production cost is 0 (just for simplicity). The firms produce identical goods and the market demand for the good is given by $D(p) = \max\{a - p, 0\}$, for $p \geq 0$, where $a > 0$. In this case the monopoly output and price are both equal to $a/2$. Assume that $0 < K < a^2/9$ (= the second stage profit each firm gets in the Nash equilibrium of the Cournot duopoly game), and that there is no discounting.

- (a) Suppose that the second-stage game is Bertrand. Describe what a strategy is for a firm in the two-stage game. Identify a subgame-perfect Nash equilibrium of this game and explain why it is subgame perfect.

Assume for the rest of the problem that the second-stage game is Cournot.

- (b) Identify a subgame-perfect Nash equilibrium of the two-stage game and explain why it is subgame perfect.
 - (c) Identify a Nash equilibrium of the two-stage game in which firm 1 gets the monopoly profit ($= a^2/4$) in the second stage. Explain why the strategy profile you identify is a Nash equilibrium and why it is not subgame perfect.
4. (Glenn Ellison) Consider a 2-player game with 2 stages: In stage 1, player 1 chooses $z \in \{0, 1, 2\}$. From this stage player 1 receives z units of money and player 2 receives $2 - z$ units of money. In stage 2, after they each observe z , they play the simultaneous move game

	L	M	R
U	x, x	$0, 0$	$-2, -2$
D	$0, 0$	$1, 1$	$-2, -2$

where x is units of money, player 1 chooses U or D, and player 2 chooses L, M, or R. The first number in each entry represents player 1's utility. Both entries are measured in *money*. (Hmm, what preference assumption ensures this?)

- (a) Sketch the extensive form. Explain what a strategy is for each player in this game and give an example.
- (b) Show that for any $x \in \mathbb{R}$ the game has a Nash equilibrium with $z = 0$.
- (c) For what values of x does the game have a unique subgame perfect Nash equilibrium (SPNE)? What is the corresponding SPNE outcome?
- (d) For what values of x is there a SPNE with $z = 0$?
- (e) Is there a SPNE outcome in which player 1's total utility ($=$ money) is less than 2?